

3.1 #s: 4, 6, 7, 11

3.2 #s: 2, 4, 8, 10

3.3 #s: 1, 3, 9, 10

Nath 3210 Homework 6

Tyler Trotter

4: Show the function $f(x) = |x|$ is continuous on all of $\mathbb{R}$.

$f(x) = |x|$ may be written as $f(x) = \begin{cases} x & \text{for } x \geq 0 \\ -x & \text{for } x < 0 \end{cases}$

So $f(x)$ can be written as two functions of the form $g(x) = x^r$ where r is a positive rational number (1).

Therefore, by Theorem 3.1.7 x and $-x$ are continuous on their respective domains (stated in the piecewise). Also, the domain spans all of $\mathbb{R}$. So $f(x) = |x|$ is continuous on all of $\mathbb{R}$. $\square$

6: Prove (d) of Theorem 3.1.5 (f/g is cont. at a , where $g(a) \neq 0$)

Using Theorem 3.1.6 & corresponding part of MLT, if f and g are cont. at a and $\{x_n\}$ is any seq. in D which converges to a , then Theorem 3.1.6 gives us $\{f(x_n)\}$ converges to $f(a)$ and $\{g(x_n)\}$ converges to $g(a)$. By part (d) of MLT, $\{f(x_n)/g(x_n)\}$ converges to $f(a)/g(a)$, assuming $g(a) \neq 0$. Therefore, f/g is continuous at a by 3.1.6. $\square$

7: Prove 3.1.11: With f and g , let $a \in D_{f \circ g}$. Then $f \circ g$ is continuous at a if g is continuous at a and f is continuous at $g(a)$.

We begin with the given if f is cont. at $g(a)$ and g is cont. at a . So, $\forall \epsilon > 0 \exists \delta_1 > 0$ s.t. $|g(x) - g(a)| < \delta_1 \Rightarrow |f(g(x)) - f(g(a))| < \epsilon$. Since g is continuous, we have $\forall \epsilon > 0, \exists \delta_2 > 0$ s.t. $|x - a| < \delta_2 \Rightarrow |g(x) - g(a)| < \delta_1$. So we have:

$$|x - a| < \delta_2 \Rightarrow |g(x) - g(a)| < \delta_1 \Rightarrow |f(g(x)) - f(g(a))| < \epsilon.$$

which is to say $f \circ g$ is continuous at a . $\square$

3.1 #s: 11

3.2 #s: 2, 4, 8, 10

3.3 #s: 1, 3, 9, 10

Homework 6, cont.

- 11: Prove that the function $f(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ is not cont. at 0.

Using the alternate characterization of continuity:
 f is cont. at 0 iff whenever $\{x_n\}$ is a seq. in D which converges to a , then $\{f(x_n)\}$ converges to $f(a)$.

We take $x_n = 1/x$, so $x_n \rightarrow 0$. However, $f(x_n)$ is defined as $\sin(x_n)$ for $x \neq 0$ and 0 for $x = 0$. Since $\{\sin(1/x)\}$ does not converge to a number as $x \rightarrow 0$, then certainly $\{f(x_n)\}$ does not converge to 0. So f is not continuous at 0. $\blacksquare$

- 2: Prove that if f is a continuous function on a closed, bounded interval I and if $f(x)$ is never 0 for $x \in I$, then there is a number $m > 0$ s.t. $f(x) \geq m$ for all $x \in I$ or $f(x) \leq -m$.

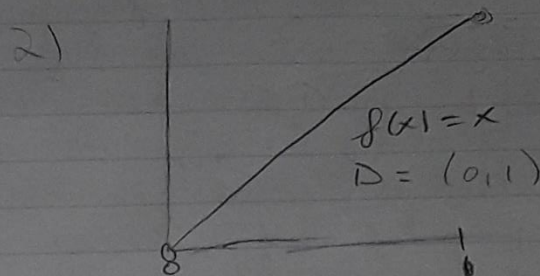
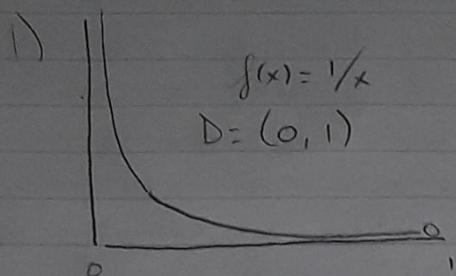
By the Intermediate Value Theorem, supposing there were a number $m < 0$ s.t. $f(x) \geq m$, $f(x)$ would be 0 for some $a \in I$. However, since $f(x)$ is never 0, clearly there exists an $m > 0$ s.t. $f(x) \geq m$ for all $x \in I$.

The statement $m \geq 0$ s.t. $f(x) > m$ would also be true. Likewise $m > 0$ is allowed because as opposed to $m \geq 0$ because the function is continuous and the function is closed and bounded. That implies f is defined at its endpoints or else f would not be continuous. So, we have on its domain a max and min value. Thus, since $f(x) \neq 0 \forall x \in I$, that min, call it l , is greater than 0. So, we may find such an m say by $0 < \frac{l}{2} = m < l$. $\blacksquare$

3.2 Hs: 4, 8, 10

3.3 Hs: 1, 3, 9, 10

2.4: Find an example of a function which is cont. on a bounded (but not closed) interval I but is not bounded. Then find an example of a function which is continuous and bounded on a bounded interval I but does not have a maximum value.



1) $f(x) = 1/x$ on the open interval $(0, 1)$ is continuous, proved in class, and is unbounded in the range.

2) $f(x) = x$ on the open interval $(0, 1)$ is continuous, 3.1.7, has a bounded range, $(0, 1)$, but has no max value because for any "max", $f(x) = (\text{"max"} + 1)/2$ that is also a value in the range $\frac{1}{2}$ on the domain.

3.2 #8, 10

3.3 #1, 3, 9, 10

Homework 6, cont

8: Show that if f and g are continuous functions on the interval $[a, b]$ such that $f(a) < g(a)$ and $g(b) < f(b)$, then there is a number $c \in (a, b)$ such that $f(c) = g(c)$.

Define a new function $h(x) = f(x) - g(x)$. We know $f(a) < g(a)$, so $f(a) - g(a) < 0$. Likewise, $g(b) < f(b)$, so $f(b) - g(b) > 0$. Since f and g are continuous on their domain, $[a, b]$, it follows that the composition that defines $h(x)$ is also continuous on that domain. So, by IVT applied to $h(x)$, where we assume $a < b$ and $h(a) < 0 < h(b)$. So, we know there exists a point where $h(c) = 0$, which is the point where $f(c) = g(c)$. ■

10: Use the Intermediate Value Theorem to prove that, if $n \in \mathbb{N}$, then every positive number a has a positive n^{th} root.

Consider the function $f(x) = a^{1/n}$, where $a \geq 0$. Fix an $n \in \mathbb{N}$. Then $f(x) = a^{1/n}$, $a^{1/2}$, $a^{1/3}$, ... Each of these are continuous functions across their domain $[0, a]$ with range $[0, a^{1/n}]$ by 3.1.7 or 3.2.5, because $f(a)$ is monotone increasing on its domain, that is if $a_1 > a_2$, then $a_1^{1/n} > a_2^{1/n}$. $f(0)$ is defined to be 0 and $f(a) = a^{1/n}$ which, by virtue of it being monotonic, implies $f(a) > f(0)$ whenever $a > 0$. Using IVT, we can show the existence of a number $f(0) < f(c) < f(a)$ given $0 \leq c \leq a$. So, since we are interested in the subset of $D / \{0\} = D'$, it follows that on D' , $\exists f(c) > 0$. This function maps a number a to its n^{th} root in the range, so we have shown that the number exists and is positive. ■

3345: 1, 3, 9, 10

Homework 6, cont.

1: Is the function $f(x) = x^2$ uniformly continuous on $(0, 1)$?

Search: $|x-a| < \delta \Rightarrow |x^2 - a^2| = |x^2 - ax + ax - a^2| =$
 $|x(x-a) + a(x-a)| \stackrel{\Delta}{=} |x(x-a)| + |a(x-a)| = x|x-a| + a|x-a|$
 $< \delta(x+a) < 2\delta < \varepsilon \quad ; \quad \delta = \frac{\varepsilon}{2}$

Proof: Let $\varepsilon > 0$. Choose $\delta = \frac{\varepsilon}{2}$. If $x, a \in (0, 1)$ and $|x-a| < \delta$,
then $|f(x) - f(a)| = |x^2 - a^2| = |x+a||x-a| < 2\delta = 2 \cdot \frac{\varepsilon}{2} = \varepsilon$. $\square$

3: Is the function $f(x) = x^2$ uniformly continuous on $(0, \infty)$? Justify.

Suppose FSOC that $f(x) = x^2$ were uniformly continuous on $(0, \infty)$. Then $\forall \varepsilon > 0 \exists \delta > 0$ s.t. $|x-a| < \delta \Rightarrow |x^2 - a^2| < \varepsilon$.

However, consider $\varepsilon = 1$. Then, suppose $a > x$ s.t. $a = x + \frac{\delta}{2}$.

Then, $|x^2 - (x + \frac{\delta}{2})^2| < 1$. However, this is the same as

$$|x^2 - (x^2 + \delta x + \frac{\delta^2}{4})| = |\delta x + \frac{\delta^2}{4}| < 1. \text{ Since } x \in (0, \infty),$$

clearly this is a contradiction for some large x . So, $f(x) = x^2$
is not uniformly continuous on $(0, \infty)$. $\square$

9: Is the function $g(x) = \sin(\frac{1}{x})$ continuous on $(0, 1)$? Is it
uniformly continuous?

Consider functions $h(x) = \sin(x)$ and $g(x) = \frac{1}{x}$. Then
 $f(x) = h(g(x))$. Since both h and g are continuous
on $(0, 1)$, so too is $f(x)$. $\square$

Now, suppose FSOC $g(x)$ were uniformly continuous on $(0, 1)$. Then
 $\forall \varepsilon > 0 \exists \delta > 0$ s.t. $|x-a| < \delta \Rightarrow |\sin \frac{1}{x} - \sin \frac{1}{a}| < \varepsilon \quad \forall x, a \in (0, 1)$.

However, consider $\varepsilon = 2$. Then consider $\sin \frac{1}{x} = -1$ and $\sin \frac{1}{a} = 1$.

So, x and y get arbitrarily close to 0, then choose $0 < x,$
 $a < \delta$. This implies $|x-a| < \delta$, but $|f(x) - f(a)| = |-1 - 1| = 2$.

This is a contradiction. So $g(x) = \sin(\frac{1}{x})$ is not uniformly
continuous on $(0, 1)$. $\square$

3.3 #5: 10

Homework 6, cont.

10: Is the function $f(x) = x \sin \frac{1}{x}$ uniformly continuous on $(0,1)$? Justify your answer.

Scratch: $|x-a| < \delta$ implies $|x \sin \frac{1}{x} - a \sin \frac{1}{a}| < \epsilon$.
 $x = \pi k$ $a = \pi(k+1)$

Proof: Let $\epsilon > 0$. Choose $\delta = \frac{\epsilon}{2}$. Then $\forall x, a \in (0,1)$,
 $|x-a| < \delta$ implies $|x \sin \frac{1}{x} - a \sin \frac{1}{a}| \stackrel{\text{triangle}}{=} |x \sin \frac{1}{x} - a \sin \frac{1}{x} + x \sin \frac{1}{x} - a \sin \frac{1}{a}|$
 $= |x-a| |\sin \frac{1}{x} - \sin \frac{1}{a}| + |x \sin \frac{1}{x} - a \sin \frac{1}{a}|$ is at most 2
because $\sin \frac{1}{x}$ may be -1 and $\sin \frac{1}{a}$ may be 1 so $|-1-1|=2$.
So, $|x-a| |\sin \frac{1}{x} - \sin \frac{1}{a}| < \frac{\epsilon}{2} \cdot 2 = \epsilon$. $\square$